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6

Approximate Solution Methods

6.1 INTRODUCTION

Although we've learned a lot in the preceding pages, it should also be clear that many problems are still beyond our analytical abilities. Anything that is more complex than a group of lumped spring-mass systems or a simple continuous system, such as a string, is currently beyond our skills. The simple addition of a nonuniform mass to a bar or a spatially varying stiffness in a beam makes the problem too difficult for closed-form analysis. These are the problems that we now address. Our solutions will not be exact, but merely approximate. Luckily, since we'll be able to create these approximate solutions to whatever degree of accuracy we desire, they'll meet all our needs. With the techniques to be introduced in the following pages, you'll be able to analyze quite general systems efficiently and accurately.

6.2 LUMPED APPROXIMATIONS

This section is probably the shortest in the book. The main focus of this chapter is on so-called modal approximations, i.e., methods that presume the existence of distributed modes for a structure. In the

opposite view, one would pay no attention to whether distributed modes might exist but instead would break the problem into little chunks. The finite element method is a well-known approach that takes this second point of view. In this approach the actual continuous body is broken into simpler units (elements) for which displacement/stress relations can be easily calculated. These elements are then joined together in a mesh to approximate the original system. The technique is very widely used and is usually taught as a separate course at the first-year graduate level. Many books are available that are devoted solely to the subject.

A different approach is to forget continuous elements entirely and try to approximate the continuous system by a collection of springs and masses (and perhaps dampers). In this way the system would look like an MDOF lumped system, something we already know how to solve. This approach will be the focus of this section.

Here's how you might proceed for a particular example. Let's assume you're examining a uniform bar. You might decide to approximate the distributed mass by n equal lumped masses located along the length of the bar. You could find the springs that connect the masses by determining the static deflection of the bar under a fixed load at the free end (the other end being restrained) and then determine what combination of discrete springs would yield this overall stiffness. As the number of masses and springs are increased, the quality of the approximation would improve, assuming you've been reasonable in assigning the mass and spring values.

The advantage of this approach is that it can be used to analyze quite general systems. The complexity of the continuous system isn't a big issue except insofar as it will generate a large number of lumped masses (which of course impacts the ease of solution).

Example 6.1

Problem Use a lumped approach to determine the first two natural frequencies for a uniform tensioned string (fixed-fixed boundary conditions). Compare the results to the exact analytical predictions. Let the length l of the string be .9 m, the total mass m_t be .05 kg, and the tension T be 1000 N/m.

Solution We'll first divide the mass of the string into n equal portions. Thus, instead of having a string of length .9 m we'll have n sections, each $\Delta = \frac{.9}{n}$ m long. All these sections will do is contribute a stiffness to the problem—they'll have no mass of their own. Between each pair of string sections we'll put a lumped mass. If we imagine each mass as a little round ball, we can view each string section as having half a ball on each end. If this is the case, then to maintain consistency we'd have to put a half-ball of mass at each of the string end points. Of course, these lumped masses won't move, since they're fixed in place owing to the boundary conditions. Nonetheless it's important to include them, since ignoring them would move onto the interior of the string than should be there more mass. This model is shown in Figure 6.1.

Counting the two half masses at the end points, we have a total of $n + 1$ complete "balls," each of mass $m = \frac{m_t}{n+1}$ kg. Since the string sections are massless, they'll always be line segments, and we can therefore determine the forces acting on each mass element in a straightforward way. If we concentrate our attention on the ith mass element, we can see that the equations of motion are given by

$$m\ddot{y}_i = \left(\frac{y_{i+1} - y_i}{\Delta}\right)T - \left(\frac{y_i - y_{i-1}}{\Delta}\right)T$$

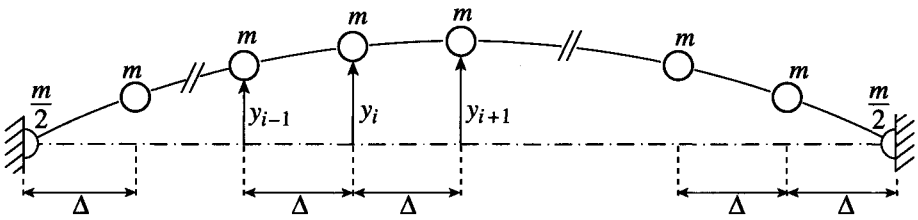

Figure 6.1 Lumped mass approximation of a tensioned string

We've used the approximation that the component of vertical force due to the tension is proportional to the angular deflection (small-angle approximation) and that the angle is approximated by the difference between the vertical deflections of two neighboring masses divided by the distance between them.

Rearranging gives us

$$m\ddot{y}_i - \frac{T}{\Delta}y_{i-1} + \frac{2T}{\Delta}y_i - \frac{T}{\Delta}y_{i+1} = 0$$

This can be expressed in matrix form as

$$\begin{bmatrix} m & 0 & 0 & \cdots \\ 0 & m & 0 & \cdots \\ 0 & 0 & m & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{Bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \\ \ddot{y}_3 \\ \vdots \end{Bmatrix} + \begin{bmatrix} \frac{2Tn}{l} & -\frac{Tn}{l} & 0 & \cdots \\ -\frac{Tn}{l} & \frac{2Tn}{l} & -\frac{Tn}{l} & \cdots \\ 0 & -\frac{Tn}{l} & \frac{2Tn}{l} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ \vdots \end{Bmatrix}$$

Dividing all the equations by m and reexpressing Δ in terms of l and n gives us

$$\begin{bmatrix} 1 & 0 & 0 & \cdots \\ 0 & 1 & 0 & \cdots \\ 0 & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{Bmatrix} \ddot{y}_1 \\ \ddot{y}_2 \\ \ddot{y}_3 \\ \vdots \end{Bmatrix} + \frac{(n+1)^2 T}{l m_t} \begin{bmatrix} 2 & -1 & 0 & \cdots \\ -1 & 2 & -1 & \cdots \\ 0 & -1 & 2 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ \vdots \end{Bmatrix}$$

If we perform an eigenanalysis on this set of equations using $n = 10$, we'll obtain the following first and second natural frequency estimates: $\tilde{\omega}_1 = 466.7$ rad/s and $\tilde{\omega}_2 = 924.0$ rad/s. The actual natural frequencies of the continuous system are given by

$$\omega_n = n\pi\sqrt{\frac{T}{m_t l}}$$

which yields results of $\omega_1 = 468.3$ rad/s and $\omega_2 = 936.6$ rad/s. As you can easily observe, the lumped approach did a good job of approximating the continuous system's natural frequencies.

As you might surmise, this lumped approach can be used in many ways. There is nothing to stop you from using nonuniform lumped masses, for instance. Thus if you wanted each mass element to be spaced uniformly along a structure, you might vary the amount of mass at each location to accurately reflect the mass distribution in the original structure.

6.3 RAYLEIGH'S QUOTIENT

One of the most amazing people in the field of vibrations was Lord Rayleigh. He was born wealthy but decided to devote his life to science (a pretty rare occurrence). At age 35, he published the *Theory of Sound* [10] in which he laid out the fundamental principles of both acoustics and vibrations. Of course, much of this material was already known, but Lord Rayleigh added a great deal of his own and was the first to put it down on paper in an easily readable form. Once you've finished the book you're now reading, you might want to find a copy of the *Theory of Sound* and read Volume One, the volume concerned with vibrations. You'll find a good bit that is very similar to what you're learning now.

Although we'll run into Rayleigh again, later in the chapter, I mention him now because we're going to discuss what's called Rayleigh's quotient. The idea behind Rayleigh's quotient is relatively straightforward. Basically, we'd like to be able to find a system's natural frequencies without having to solve the associated eigenvalue problem. Now you might think this isn't possible; it would be like getting something for nothing. However, if we already know the eigensolutions to a problem that's *close* to the one we're currently interested in, we can exploit this knowledge to our advantage. We'll consider discrete systems first and then move on to continuous ones in Section 6.5.

If we're dealing with an undamped, lumped mass system, we know that the equations of motion are of the form

$$[M]\ddot{X} + [K]X = O \tag{6.3.1}$$

We can recognize that this is an eigenvalue problem, and thus we know the solutions are given by

$$-\omega_i^2[M]X_i + [K]X_i = O \tag{6.3.2}$$

where X_i is the ith eigenvector and ω_i is the ith natural frequency of oscillation. What we'll do next is premultiply both sides of this equation by $\frac{1}{2}X_i^T$:

$$-\tfrac{1}{2}\omega_i^2 X_i^T[M]X_i + \tfrac{1}{2}X_i^T[K]X_i = O \tag{6.3.3}$$

It's worth discussing why we might want to do this at this point. For ease, let's look at a 3 DOF system (Figure 6.2). A 2 DOF system would be even easier, but we'll be looking at some characteristics that require at least three degrees of freedom in a few pages, so we'll stick with 3 DOF.

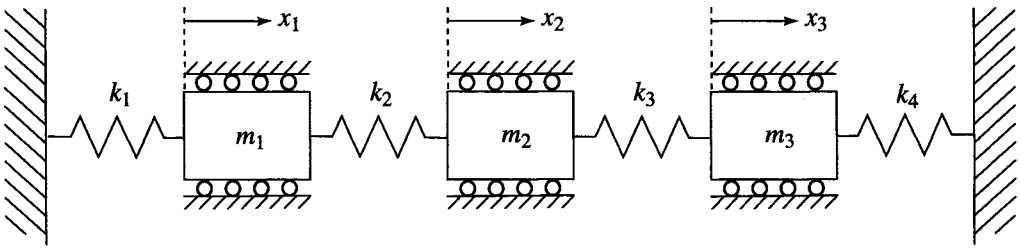

Figure 6.2 3 DOF spring mass system

We know that the potential and kinetic energies of the system can be written as

$$PE = \tfrac{1}{2}k_1x_1^2 + \tfrac{1}{2}k_2(x_2 - x_1)^2 + \tfrac{1}{2}k_3(x_3 - x_2)^2 + \tfrac{1}{2}k_4x_3^2 \tag{6.3.4}$$

and

$$KE = \tfrac{1}{2}m_1\dot{x}_1^2 + \tfrac{1}{2}m_2\dot{x}_2^2 + \tfrac{1}{2}m_3\dot{x}_3^2 \tag{6.3.5}$$

These are all quadratic terms; i.e., everything is raised to the second power. Thus we can write them in matrix form as

$$PE = \tfrac{1}{2}\{x_1 \;\; x_2 \;\; x_3\}\begin{bmatrix} k_1 + k_2 & -k_2 & 0 \\ -k_2 & k_2 + k_3 & -k_3 \\ 0 & -k_3 & k_3 + k_4 \end{bmatrix}\begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} \tag{6.3.6}$$

and

$$KE = \tfrac{1}{2}\{\dot{x}_1 \;\; \dot{x}_2 \;\; \dot{x}_3\}\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix}\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{Bmatrix} \tag{6.3.7}$$

More compactly, this is just

$$PE = \tfrac{1}{2}X^T[K]X \tag{6.3.8}$$

and

$$KE = \tfrac{1}{2}\dot{X}^T[M]\dot{X} \tag{6.3.9}$$

where $[M]$ and $[K]$ are our familiar mass and stiffness matrices. If we go one step further and ask what the energies would be if the system was oscillating at one of its natural frequencies, say ω_i, then (6.3.8) and (6.3.9) would become

$$PE = \tfrac{1}{2}X_i^T[K]X_i\cos^2(\omega_i t) \tag{6.3.10}$$

$$KE = \tfrac{1}{2}\omega_i^2X_i^T[M]X_i\sin^2(\omega_i t) \tag{6.3.11}$$

where X_i is the ith eigenvector and $X = X_i\cos(\omega_i t)$ describes the motion.

Both these energies are time-varying functions that oscillate between zero and a maximum value. As a glance at (6.3.10) and (6.3.11) reveals, the maximum value of the potential energy is $\frac{1}{2}X_i^T[K]X_i$, while the maximum kinetic energy is $\frac{1}{2}\omega_i^2X_i^T[M]X_i$. Using an overbar to represent the maximum, we therefore have

$$\overline{PE} = \tfrac{1}{2}X_i^T[K]X_i \tag{6.3.12}$$

and

$$\overline{KE} = \tfrac{1}{2}\omega_i^2X_i^T[M]X_i \tag{6.3.13}$$

We're going to need another term in a second, so we'll introduce it now. If we factor the ω^2 out of (6.3.13) we'll obtain what we'll call the *zero-frequency kinetic energy*, which we'll indicate by a zero superscript:

$$\overline{KE}^0 = \tfrac{1}{2} X_i^T [M] X_i \tag{6.3.14}$$

Now compare (6.3.12) and (6.3.13) with (6.3.3). We can see that the first term of (6.3.3) is just the maximum kinetic energy of the system and the second is the maximum potential energy. So what we've done by premultiplying by X_i^T is to change what were force terms into energy terms. If we now divide (6.3.3) by $X_i^T [M] X_i$ we'll have

$$\omega_i^2 = \frac{\frac{1}{2} X_i^T [K] X_i}{\frac{1}{2} X_i^T [M] X_i} \tag{6.3.15}$$

What this tells us is that the square of a system's natural frequency is *always* equal to the ratio of the system's maximum potential energy to the system's zero-frequency kinetic energy. Since we're concerned with the ratio of the two energies, the $\frac{1}{2}$ factors aren't a problem; we can just as well factor them out,

$$\omega_i^2 = \frac{X_i^T [K] X_i}{X_i^T [M] X_i} \tag{6.3.16}$$

The more general form of (6.3.16), one that simply depends on an arbitrary input vector, is given by

$$R_Q(X) = \frac{X^T [K] X}{X^T [M] X} \tag{6.3.17}$$

where R_Q stands for, not surprisingly, Rayleigh's quotient. Any given problem has an infinite number of values for the Rayleigh's quotient, since changing X will change $R_Q(X)$. Notice that simply scaling the input vector doesn't alter Rayleigh's quotient; i.e., $R_Q(X) = R_Q(nX)$, where n is any constant. This is in keeping with our notions of eigensolutions—and the fact that the eigenvectors can be determined only up to a multiplicative constant. It is important to remember that Rayleigh's quotient will produce a continuum of values, depending upon X. For the special cases of the input X being equal to an eigenvector, Rayleigh's quotient will equal the square of the natural frequency associated with that eigenvector.

Equation (6.3.17) is very significant, not just for our discussion of Rayleigh's quotient but for our later work in approximation methods. Basically, it's a way of getting the eigenvalues for our problem without solving an eigenvalue problem. If (and that's a *big* if right now) someone hands us the actual eigenvectors of our problem, then simply substituting them into (6.3.17) will give us the associated eigenvalue (or equivalently, the square of the natural frequency).

As it now stands, this may not seem like a really big deal. First, you have to wonder how often someone will conveniently hand you the system eigenvectors. And if you don't have the exact eigenvectors to work from, but only an approximation of them, who's to say that (6.3.17) won't just

give an inaccurate estimate of the natural frequencies? Well, the good news is that there's a bit more to (6.3.17) than there initially appears to be. In the first place, design invariably involves an iterative process in which a system is "tweaked" in various ways to improve its performance. Generally, you know a good deal about the system before the "tweaking" starts. In a similar vein, improvements of an existing product often consist of small modifications to the basic unit, modifications that are, again, meant to improve performance in some way. Here also, the basic system is well understood before modifications begin. Thus there is a very good chance that the system eigenmodes and natural frequencies have already been catalogued and are ready for use in later analyses. Since the changes to the existing system are generally not too large, it seems reasonable that the eigenmodes of the system will not change drastically. Therefore the old eigenmodes make very reasonable starting estimates for the modified system's eigenmodes.

So let's say that someone gives you (or you estimate for yourself) an approximation to the ith eigenvector of the modified system. You then take this approximation and use it in (6.3.17). You obviously won't obtain ω_i^2, because the only way to get that would be to have used the *exact* ith eigenvector. But, what you *will* obtain is an extremely good estimate of ω_i^2. In fact, the error in the estimate of ω_i^2 will be far less than the error in your approximation of the ith eigenvector. That's the power of (6.3.17); it gives a good quality estimate for the natural frequency even if your input eigenvector estimate isn't very good.

Example 6.2

Problem Let's look at the system shown in Figure 6.3 to illustrate how well Rayleigh's quotient can work. Figure 6.3*a* represents our original system. We then impose some changes in the system based on design considerations and wind up with Figure 6.3*b*. Use Rayleigh's quotient to determine how the natural frequencies have changed .

Solution The original system has the following mass and stiffness matrices

$$[M] = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 10 \end{bmatrix}, \quad [K] = \begin{bmatrix} 200 & -100 & 0 \\ -100 & 200 & -100 \\ 0 & -100 & 200 \end{bmatrix}$$

which support the eigenvectors

$$X_1 = \begin{Bmatrix} .5000 \\ .7071 \\ .5000 \end{Bmatrix}, \quad X_2 = \begin{Bmatrix} .7071 \\ .0000 \\ -.7071 \end{Bmatrix}, \quad \text{and} \quad X_3 = \begin{Bmatrix} .5000 \\ -.7071 \\ .5000 \end{Bmatrix}$$

with $\omega_1^2 = 5.858$ rad/s, $\omega_2^2 = 20.00$ rad/s, and $\omega_3^2 = 34.14$ rad/s.

The modified system (indicated by a hat) has mass and stiffness matrices

$$[\hat{M}] = \begin{bmatrix} 11 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 12 \end{bmatrix}, \quad [\hat{K}] = \begin{bmatrix} 175 & -90 & 0 \\ -90 & 185 & -95 \\ 0 & -95 & 185 \end{bmatrix}$$

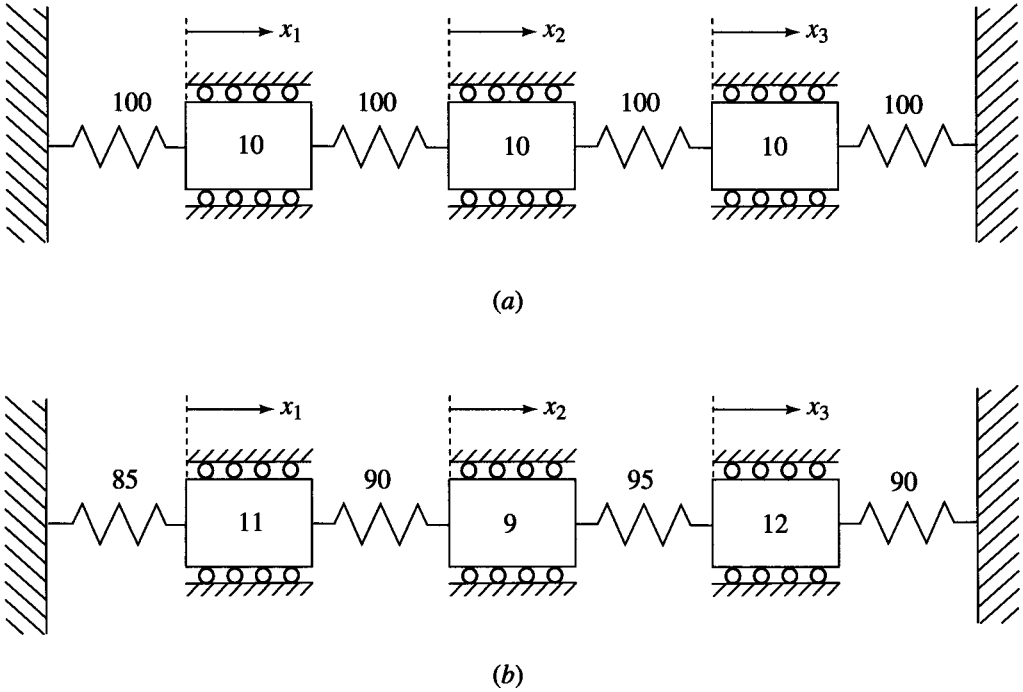

Figure 6.3 Two 3 DOF spring-mass system: (*a*) unperturbed and (*b*) perturbed

which support the eigenvectors

$$\hat{X}_1 = \begin{Bmatrix} .5130 \\ .6831 \\ .5198 \end{Bmatrix}, \quad \hat{X}_2 = \begin{Bmatrix} .7308 \\ .0216 \\ -.6823 \end{Bmatrix}, \quad \text{and} \quad \hat{X}_3 = \begin{Bmatrix} .4315 \\ -.8064 \\ .4045 \end{Bmatrix}$$

with $\hat{\omega}_1^2 = 5.014$ rad/s, $\hat{\omega}_2^2$=15.67 rad/s, and $\hat{\omega}_3^2 = 31.20$ rad/s.

Keep in mind that this problem was designed to be simple, and so there's no real reason to use Rayleigh's quotient; it isn't a big deal to solve the problem analytically. That's why both solutions have been solved exactly—so we can see how well Rayleigh's quotient works. Notice that the eigenvalues of the two problems (the square of the natural frequencies) have changed substantially, as have the eigenvectors.

What we'll do now is use the original eigenvectors as estimates of the new eigenvectors and utilize Rayleigh's quotient to determine the new natural frequencies. Thus, for the first natural frequency estimate (which we'll denote by a tilde), we have

$$\tilde{\omega}_1^2 = R_Q(X_1) = \frac{X_1^T[\hat{K}]X_1}{X_1^T[\hat{M}]X_1}$$

We're trying to estimate the new natural frequency by using a good guess for the eigenvector and our knowledge of what the new mass and stiffness matrices actually are. Using the foregoing values, what

we'll find is that

$$\tilde{\omega}_1^2 = 5.043 \text{ rad/s}$$

If we go ahead and determine the estimates of the second and third natural frequencies, using X_2 and X_3, respectively, we'll obtain

$$\tilde{\omega}_2^2 = 15.65 \text{ rad/s} \quad \text{and} \quad \tilde{\omega}_3^2 = 30.57 \text{ rad/s}$$

Compare these results with the squares of the actual natural frequencies (5.014, 15.67, and 31.20 rad/s). The results are fantastically good! 5.043 rad/s is off only by half a percent from 5.014 rad/s, 15.65 rad/s is off by .1%, and 30.57 rad/s is off by 2%. Rayleigh's quotient gave us absolutely excellent estimates of the modified natural frequencies and did it using eigenvectors that were noticeably off from the actual eigensolutions.

Having seen that Rayleigh's quotient does this nifty job of producing answers that are be than the data we feed in, it's time to find out why this happens. The answer has a lot to do with what's happening in a geometric sense with the system's eigenstructure. This is a bit complicated to visualize, so we'll break the problem down and look at bits of it individually before putting it all together.

First let's see how Rayleigh's quotient varies as we use different amounts of a particular eigenvector. Figure 6.4 shows Rayleigh's quotient for our original system (6.2). In these plots we're using the actual eigenvectors of the system and simply changing how much is used. As you can see, the plots are flat; it makes no difference whether we use a lot or a little. This is completely in line with our notion of eigenvectors. All that matters are the relative displacements, not the actual magnitudes.

Now we'll go on to something more interesting. In Figure 6.5, we're again plotting Rayleigh's quotient. However this time our estimated vector consists of one eigenvector (X_1) plus a varying amount of another. More specifically, in Figure 6.5*a* we're letting our estimated vector equal $X_1 + a_2X_2$ while keeping a_3 (the amount of the third eigenvector) at zero. When a_2 is zero, Figure 6.5*a* shows that R_Q gives us the correct value for ω_1^2, namely, 5.858. We expect the graph to exactly equal the square of the first natural frequency when a_2 equals zero, and it does. The real question is what happens if a_2 isn't zero. As we add more and more of a contamination vector (by letting a_2 grow as in Figure 6.5*a*); the R_Q estimate increases. Near the origin, the plot looks quadratic, and it levels off for high values of a_2. The same qualitative behavior is seen in Figure 6.6*b* where we're contaminating X_1 with X_3.

This is behavior is always present; the *lowest* value that Rayleigh's quotient can produce is the square of the first natural frequency. No input vector can produce a lower output. Therefore one extremely inefficient way to determine the modified system's first natural frequency would be to try every conceivable trial vector. The one giving you the lowest value for Rayleigh's quotient will be the first eigenvector. Of course, nobody would ever use such an approach, but at least we know we could if all else failed.

The next set of plots (Figure 6.6) shows how Rayleigh's quotient varies when we let the estimated vector equal $X_3 + a_1X_1$ (Figure 6.6*a*) and then let it equal $X_3 + a_2X_2$ (Figure 6.6*b*). We see that the plots have maximums when the estimated vector is simply X_3: i.e., the value decreases as we add

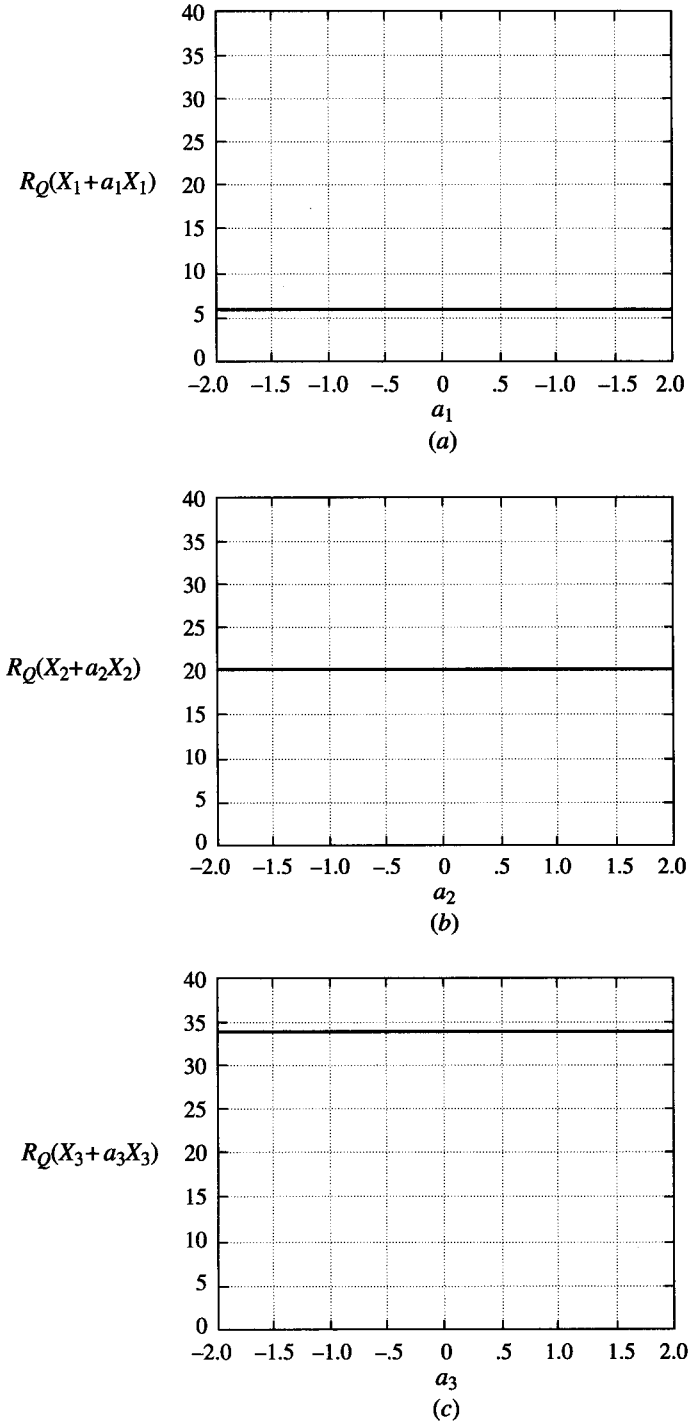

Figure 6.4 Behavior of Rayleigh's quotient about each eigenvector when perturbed by that eigenvector

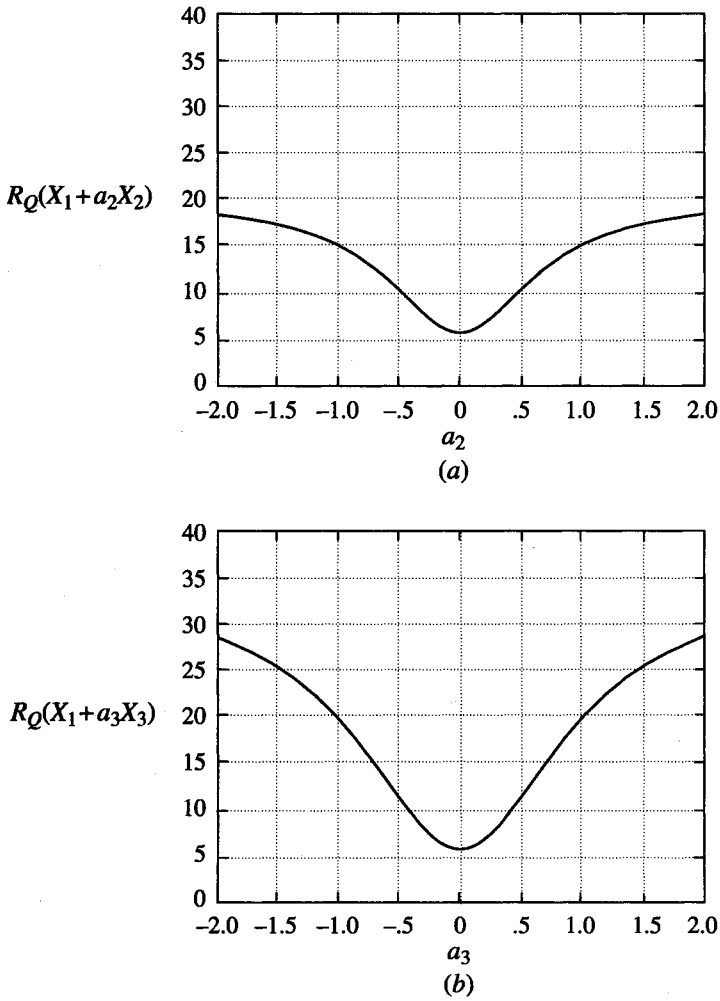

Figure 6.5 Behavior of Rayleigh's quotient about the first eigenvector

components of X_1 or X_2 to it. Thus we have another nugget of information, namely, that the *highest* possible value for Rayleigh's quotient is equal to the square of the largest natural frequency (in this case, the third natural frequency).

Now it's time for the weirder result. The first two cases were nicely behaved. Contaminating the first eigenvector (the one associated with the *lowest* natural frequency) with any other eigenvector simply made Rayleigh's quotient increase, while adding some other eigenvector to the *last* eigenvector (the one associated with the highest natural frequency) caused Rayleigh's quotient to decrease. These are the two simplest cases—contaminating the first or the last eigenvector. The behavior is more complex for any other eigenvector. Of course, in this example there is only one other eigenvector to consider, since it's only a three-mass system. But for more complex systems, all eigenvectors

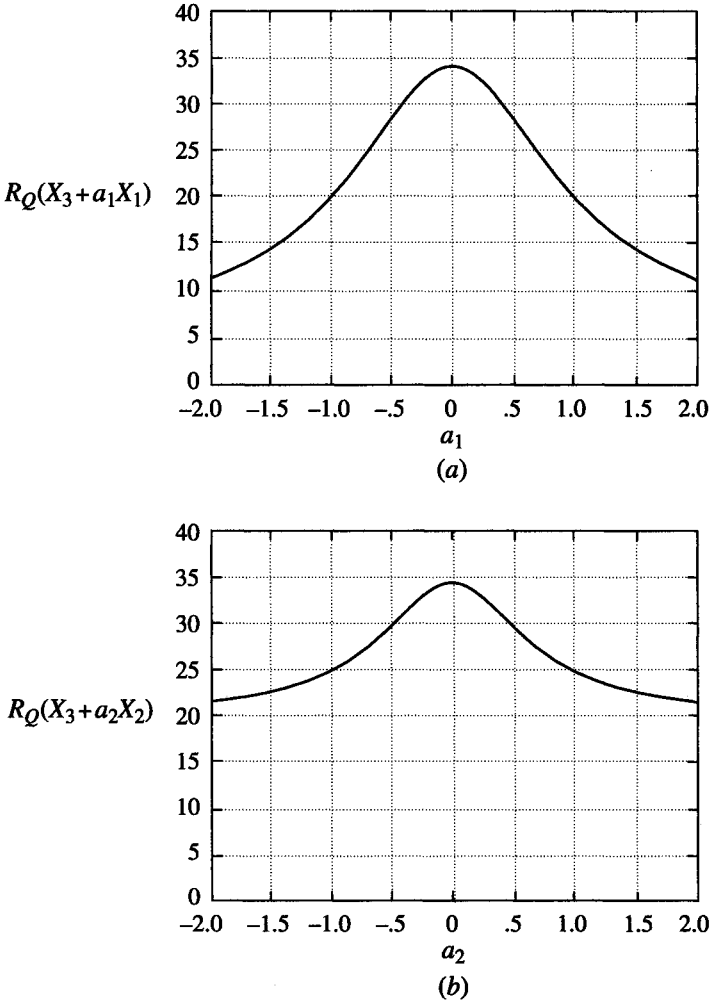

Figure 6.6 Behavior of Rayleigh's quotient about the third eigenvector

other than the ones corresponding to the lowest or highest natural frequency will display the kind of behavior we'll see next.

Figure 6.7 shows how Rayleigh's quotient changes as we contaminate the second eigenvector with either the first eigenvector ($X_2 + a_1X_1$: Figure 6.7*a*) or the third eigenvector ($X_2 + a_3X_3$: Figure 6.7*b*). We can see from these plots that Rayleigh's quotient *decreases* when we add some of the first eigenvector (X_1) but *increases* if we add some of the third eigenvector (X_3). This actually makes sense if you think about what Rayleigh's quotient does. As we've seen, Rayleigh's quotient spits out a scalar that corresponds to the system's natural frequencies if the input is an eigenvector. We know that the second natural frequency is less than the third and more than the first. By adding some of X_3 to our estimated vector, we're reorienting the estimated vector toward the third eigenvector.

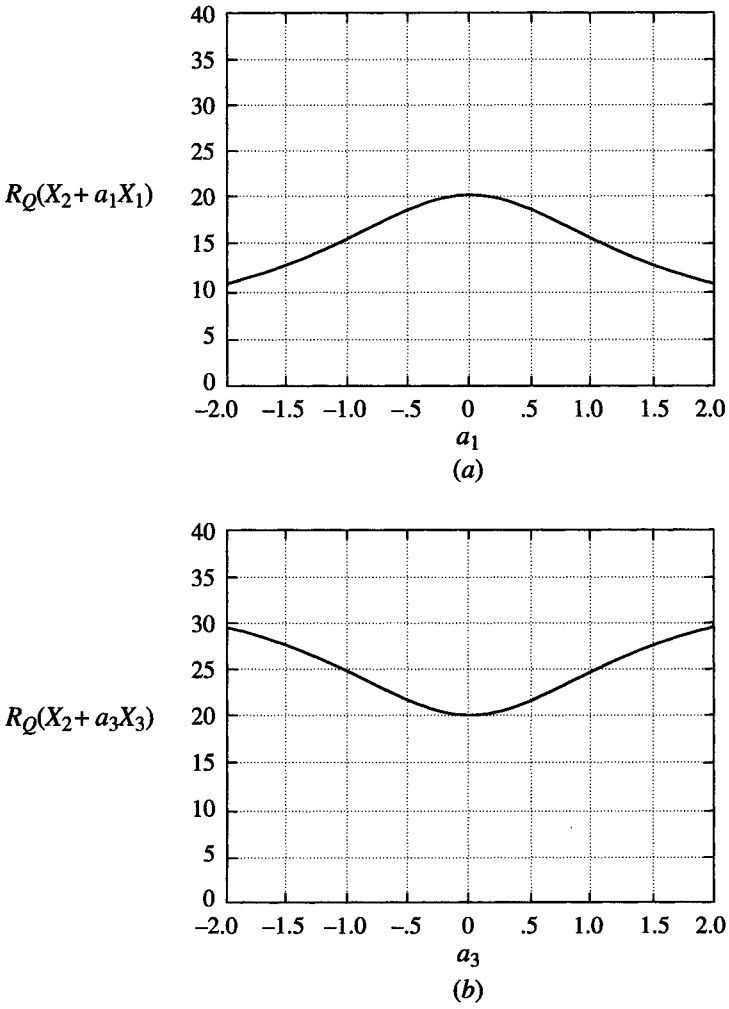

Figure 6.7 Behavior of Rayleigh's quotient about the second eigenvector

If we add a lot of X_3, then the second becomes trivial in comparison, thus making the estimated vector approximate X_3. And Rayleigh's quotient for the third eigenvector is *bigger* than for the second. So we should expect Rayleigh's quotient to increase as we add more and more of X_3. Otherwise, how would the result smoothly increase to the higher value?

The same logic holds when we add some of X_1. In this case Rayleigh's quotient will end up smaller as we get more and more of the first eigenvector involved, and so we'd expect it to decrease steadily toward this lower value.

Of course, it's more complicated if we add a little of *both* remaining eigenvectors to an initial eigenvector. When the initial eigenvector is the first or the last, we still get a minimum or maximum (Figure 6.8), since both lead in the same direction (toward a lower or higher value, depending on

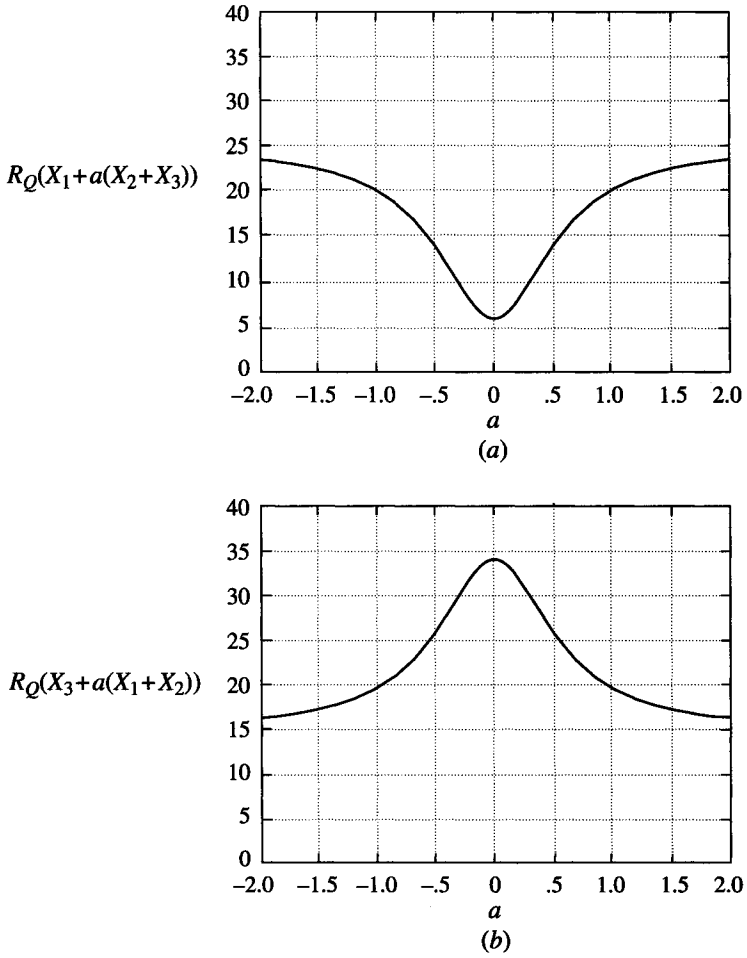

Figure 6.8 Behavior of Rayleigh's quotient about the first and third eigenvectors when perturbed by a combination of both remaining eigenvectors

whether the initial eigenvector is the highest or lowest one). But for the case of the second eigenvector we've got two competing trends. Adding some X_1 will cause Rayleigh's quotient to decrease, but adding some X_3 will cause it to increase. Therefore the overall result will depend upon how much of each is added. As an example, Figure 6.9 illustrates how Rayleigh's quotient will vary as equal amounts of X_1 and X_3 are added to X_2. Note that the result is *no* variation: the effect of X_1 is exactly canceled by that of X_3, resulting in no change to Rayleigh's quotient.

This behavior is the same as we'd see in higher dimensional systems for all eigenvectors except those associated with the lowest and highest natural frequencies. Assuming that the natural frequencies have been ordered from lowest to highest, perturbing off X_i with some of X_j will cause the Rayleigh's quotient to increase if j is greater than i and decrease if j is less than i.

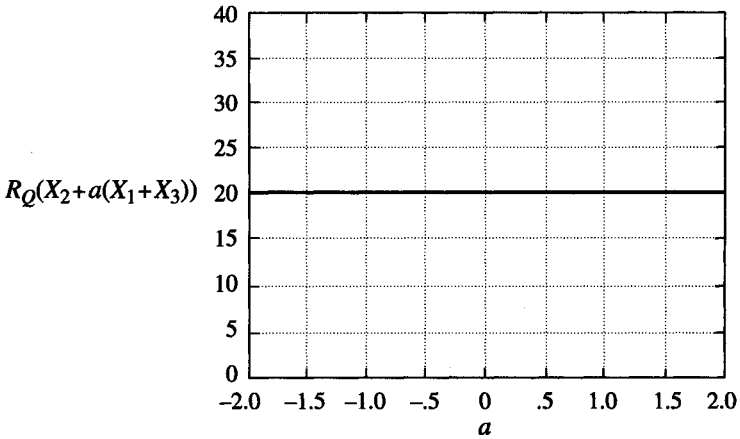

Figure 6.9 Behavior of Rayleigh's quotient about the second eigenvector when perturbed by a combination of both remaining eigenvectors

Time for the final observation. You may recall from your multivariable calculus class that when a function has a minimum, the derivative at that minimum with respect to the independent variable is zero. The same holds true at a maximum and also at inflection points. We can see this clearly in Figure 6.10. The slope of Rayleigh's quotient versus a_1, a_2, etc. is always zero at the minimum. For the case of X_2 (Figure 6.7) we still have a slope of zero with respect to the a_i's; the only thing different is that we have both a maximum and a minimum, depending upon which independent variable we're looking at. Even when we're looking at combinations of the a_i's (Figure 6.8), we still have a zero slope when the a_i's equal zero. Let's take a closer look at a particular case. Figure 6.11, an enlargement of Figure 6.5*b*, shows how Rayleigh's quotient varies when $a_2 = 0$ and $X_1 + a_3X_3$ is the input. If we're near $a_3 = 0$, then, because the slope is zero, small changes in a_3 don't appreciably alter Rayleigh's quotient. The point labeled A on the graph shows the change in Rayleigh's quotient when $a_3 = .10$. As you can see, the value doesn't change much at all from what it was when $a_3 = 0$. This is a fundamental property of minima, maxima, or inflection points. Such points are called stationary points because the dependent function is essentially stationary about them and doesn't change (much) as the dependent variable is altered.

This is the result we've been looking for! If we have a halfway decent quess for a modified system's eigenvector, then our estimated vector has a lot of the actual eigenvector in it, along with some contamination from the other eigenvectors. What we've now found out is that this contamination doesn't really alter the result very much, as long as contamination isn't excessive. Therefore pretty good guesses about the eigenvectors will give us *very good* estimates of the natural frequencies.

Having numerically demonstrated all this, it's time to look at the mathematics to see *why* Rayleigh's quotient behaves as it does. We'll start by examining (6.3.17) more closely. We already know that the eigenvectors of a system span the system's configuration space. Thus, any vector that we come up with can be represented by an appropriate combination of the system's eigenvectors. We also know from our recent work that Rayleigh's quotient is stationary about the system's eigenvectors.

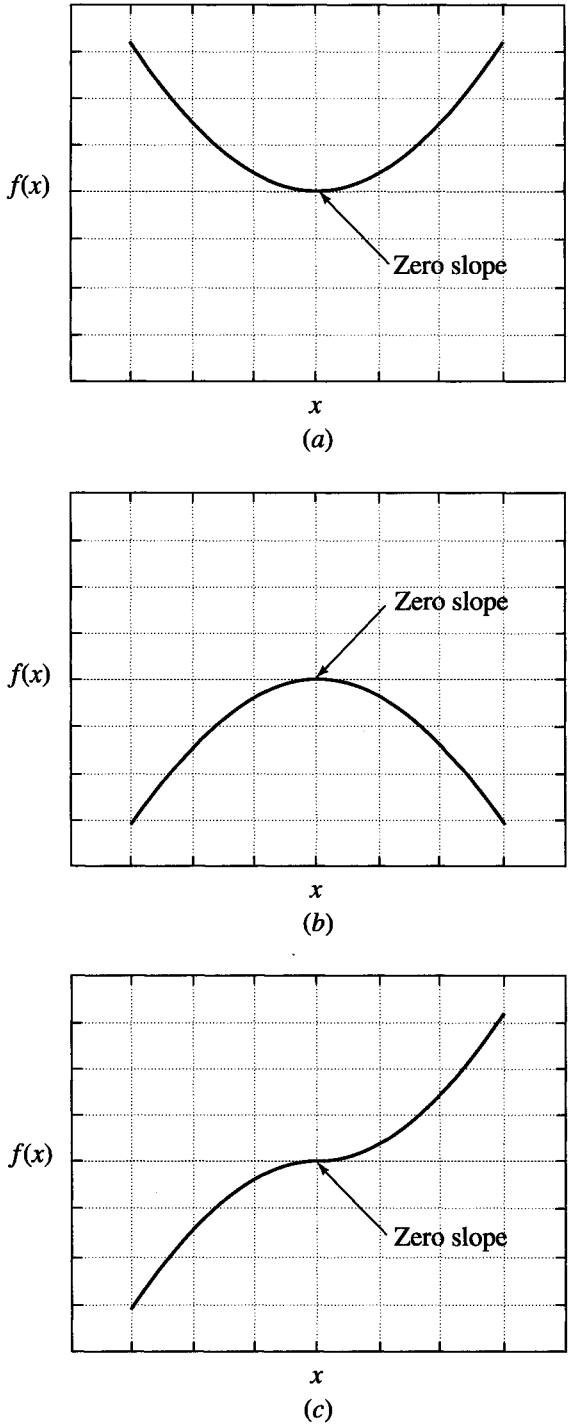

Figure 6.10 Zero-slope behavior when a function is evaluated at a minimum, a maximum, and an inflection point

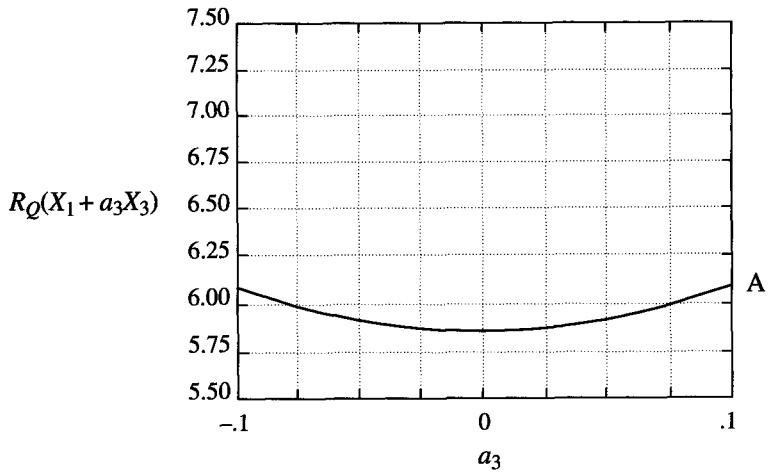

Figure 6.11 Behavior of Rayleigh's quotient about the first eigenvector (enlarged)

We'll assume that our estimated vector is given by

$$\hat{X} = a_1X_1 + a_2X_2 + \cdots + a_{j-1}X_{j-1} + X_j + a_{j+1}X_{j+1} + \cdots + a_nX_n \tag{6.3.18}$$

that is,

$$\hat{X} = \sum_{i=1}^{n} a_iX_i \tag{6.3.19}$$

where $a_j = 1$. We'll further assume that the remaining a_i's are small ($a_i \ll 1$). Thus $\hat{X}$ is mainly equal to the jth eigenvector X_j, with only small contamination from the other eigenvectors. This is the definition of a *good* estimate: it's close to, but not exactly equal to, the actual eigenvector. We don't have to pin down the a_i's exactly; it's enough to know that they're small.

Our next step is to see what Rayleigh's quotient looks like in terms of our estimate $\hat{X}$. Using (6.3.19) in (6.3.17) gives us

$$R_Q(\hat{X}) = \frac{\left(\sum_{i=1}^{n} a_iX_i^T\right)[K]\left(\sum_{i=1}^{n} a_iX_i\right)}{\left(\sum_{i=1}^{n} a_iX_i^T\right)[M]\left(\sum_{i=1}^{n} a_iX_i\right)} \tag{6.3.20}$$

For clarity, we can explicitly expand the summation out:

$$R_Q(\hat{X}) = \frac{(a_1X_1^T + a_2X_2^T + \cdots)[K](a_1X_1 + a_2X_2 + \cdots)}{(a_1X_1^T + a_2X_2^T + \cdots)[M](a_1X_1 + a_2X_2 + \cdots)} \tag{6.3.21}$$

We now need to recall that the eigenvectors of a system are orthogonal with respect to the $[K]$ and $[M]$ matrices. Therefore many of the multiplications in (6.3.21) will simply yield zero. Let's also decide that our eigenvectors have been normalized, so that

$$X_i^T[M]X_i = 1 \tag{6.3.22}$$

This step isn't actually necessary, it just makes the mathematics a bit clearer. If the vectors have been normalized in this way, then we know from Chapter 4 that

$$X_i^T[K]X_i = \omega_i^2 \tag{6.3.23}$$

If we now use the orthogonality properties on (6.3.21) we'll be left with

$$R_Q(\hat{X}) = \frac{a_1^2\omega_1^2 + a_2^2\omega_2^2 + \cdots + \omega_j^2 + \cdots a_n^2\omega_n^2}{a_1^2 + a_2^2 + \cdots + 1.0 + \cdots + a_n^2} \tag{6.3.24}$$

The denominator can be rearranged slightly, as follows:

$$R_Q(\hat{X}) = \frac{a_1^2\omega_1^2 + a_2^2\omega_2^2 + \cdots + \omega_j^2 + \cdots a_n^2\omega_n^2}{1 + a_1^2 + a_2^2 + \cdots} \tag{6.3.25}$$

Since all the a_i's are small, the denominator is in the form

$$1 + \varepsilon \tag{6.3.26}$$

where ε is small. You may recall that the inverse of $1 + \varepsilon$ is given by

$$\frac{1}{1+\varepsilon} = 1 - \varepsilon + \varepsilon^2 + \cdots \tag{6.3.27}$$

If we use this approximation in (6.3.25) we can rewrite Rayleigh's quotient as

$$R_Q(\hat{X}) = \left(a_1^2\omega_1^2 + a_2^2\omega_2^2 + \cdots + \omega_j^2 + \cdots a_n^2\omega_n^2\right)\left(1 - (a_1^2 + a_2^2 + \cdots) + (a_1^2 + a_2^2 + \cdots)^2 - \cdots\right) \tag{6.3.28}$$

All we'll need to be interested in are the terms up to second order (terms like a_1^2), and so when we expand out (6.3.28) we'll neglect all higher order terms. Therefore

$$\begin{aligned} R_Q(\hat{X}) \approx\ & \omega_j^2 + a_1^2(\omega_1^2 - \omega_j^2) + a_2^2(\omega_2^2 - \omega_j^2) + \cdots + a_{j-1}^2(\omega_{j-1}^2 - \omega_j^2) \\ & + a_{j+1}^2(\omega_{j+1}^2 - \omega_j^2) + \cdots + a_n^2(\omega_n^2 - \omega_j^2) \end{aligned} \tag{6.3.29}$$

Let's take a second to think about what we've got. What we see is that our Rayleigh's quotient is equal to the square of the jth natural frequency plus error terms. Furthermore, all the error terms are *second-order terms*! Even though our input vector had first-order errors (terms like a_1X_1), no equivalent errors are showing up in our Rayleigh's quotient approximation. If they had appeared we'd be seeing terms like $a_1\,\omega_1^2$ and $a_2\,\omega_j^2$. This is what gives us our great accuracy. The errors in the input depend on the a_i's, which are small. Well, if the a_i's are small, then the values of a_i^2 are absolutely tiny.

Equation (6.3.29) shows that our approximate value is equal to the actual eigenvalue we're seeking (the ω_j^2 term) plus some tiny errors that are proportional to the a_i^2's. Any decent guess leads to an excellent approximation, just as we've already seen.

Finally, can (6.3.29) also predict the precise kind of behavior we saw in Figures 6.5–6.9? Absolutely. Let's first consider what we get if we're looking for the first natural frequency. In this case, $j = 1$ and (6.3.29) becomes

$$R_Q(\hat{X}) = \omega_1^2 + a_2^2(\omega_2^2 - \omega_1^2) + a_3^2(\omega_3^2 - \omega_1^2) + \cdots \tag{6.3.30}$$

Since ω_1 is the lowest natural frequency, all the error terms are positive (the a_i^2 are always positive and the $\omega_i^2 - \omega_1^2$ terms are also positive, since $\omega_i > \omega_1$). Since all the error terms are positive, we see that R_Q is a minimum around the first eigenvector; all contaminating vectors can serve only to increase its value.

If we go to the other end of the spectrum and look at the highest natural frequency, then

$$R_Q(\hat{X}) = \omega_n^2 + a_1^2(\omega_1^2 - \omega_n^2) + a_2^2(\omega_2^2 - \omega_n^2) + \cdots \tag{6.3.31}$$

For this case R_Q is a maximum, since any contamination will reduce its value (the a_i^2 terms are still positive, and all the differences between natural frequencies are negative).

For any other natural frequency, we'll have some positive and some negative terms. This can be seen in (6.3.29). Whenever $i > j$, the $a_i^2(\omega_i^2 - \omega_j^2)$ terms will be positive (causing positive errors) and whenever $i < j$, the $a_i^2(\omega_i^2 - \omega_j^2)$ terms will be negative (causing negative errors).

All these characteristics can be seen in Figures 6.5–6.9. The only thing to keep in mind when looking at these figures and our mathematical results is that our entire analysis of Rayleigh's quotient has been based on input vectors that are close to the actual eigenvectors. The results are therefore valid only for small a_i, as stated earlier. The plots in Figures 6.5–6.9 hold for both small and large values of a_i. Thus the analysis and figures will match for small values of a_i but diverge as the a_i get too large. For instance, Figure 6.5*b* shows how Rayleigh's quotient behaves for the first eigenvector, with some contamination from the third. Our mathematical analysis says that the plot should look like

$$R_Q(\hat{X}) = \omega_1^2 + a_3(\omega_3^2 - \omega_1^2) \tag{6.3.32}$$

which it does, for small values of a_3. However, as a_3 increases, the plot levels off, approaching ω_3^2, not a surprising result because X_3 becomes the dominant component of $\hat{X}$ as a_3 increases. Figure 6.12 superimposes both these solutions, showing how the two systems behave similarly when a_3 is small and diverge as the magnitude of a_3 grows.

One additional observation comes directly from looking at Rayleigh's quotient. Since Rayleigh's quotient is equal to $\frac{X^T[K]X}{X^T[M]X}$, we can see that increasing $[K]$, while holding $[M]$ fixed, will increase the resulting frequency estimates. Thus increasing $[K]$, which means increasing the stiffness in the system, results in higher natural frequencies. Similarly, increasing the mass of the system (leading to larger values of $[M]$) serves to decrease the natural frequencies $\left(\text{since Rayleigh's quotient is proportional to } \frac{1}{X^T[M]X}\right)$. This is in line with our intuition and with results we've already seen.

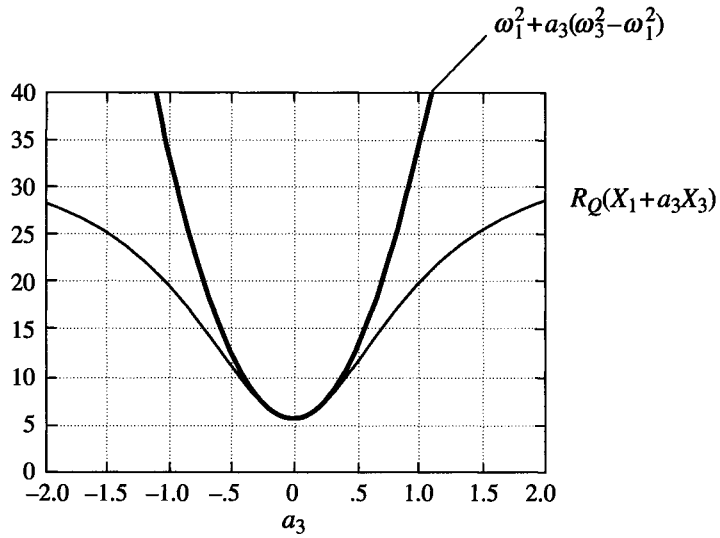

Figure 6.12 Comparison of Rayleigh's quotient and a second-order polynomial approximation

6.4 RAYLEIGH-RITZ METHOD: DISCRETE SYSTEMS

Rayleigh's quotient is a powerful tool for estimating system natural frequencies, but we can go quite a bit further to improve our results. The first person to realize this was Walter Ritz, and for this reason, the method is called the Rayleigh-Ritz method. The real power of the Rayleigh-Ritz method lies in how it allows us to analyze complicated continuous problems. To illustrate how the method works, however, we're first going to confine our attention to finite-dimensional systems. This will make it much easier to grasp what the mathematics is doing for us. Only after we've mastered a finite-dimensional application of the method will we move on to the more physically meaningful continuous problems.

Okay, back to the method. Ritz looked at the results we've just finished deriving and asked himself if there was any way to extract even more information from them. His logic went something like this:

> Rayleigh's quotient gives us a good way of estimating a system's fundamental frequency. If I use a pretty good estimate for the first eigenvector, I'll get an excellent approximation. But I'd like to know more than just the fundamental frequency; I'd like several natural frequencies, and I'd like to know their associated eigenvectors. Is there a way to use more than one estimated vector, and mix these estimated vectors around in an appropriate way, so that I end up with estimates for several natural frequencies and several eigenvectors? And can I work it so that, as I use more and more estimated vectors, my results get better and better?

Well, what Ritz decided, after a bit of rumination, was that the fundamental reality of Rayleigh's quotient was its *stationarity* when perturbed around any of the system's actual eigenvectors.

(We studied this fact in detail in Section 6.3.) So Ritz decided to generalize the concept of an estimated vector. Rather than use just a single vector as an estimated vector, Ritz used several. Thus, he first decided on a *set* of trial vectors, $\hat{X}_1$, $\hat{X}_2$, ... Keep in mind that (unless you're real lucky in choosing your trial vectors), these trial vectors are *not* the eigenvectors. Each $\hat{X}_i$ is simply your best estimate of the system's eigenvectors. They might be good estimates or they might be lousy. What we'll see is that, either way, you'll get good results from the Rayleigh-Ritz method as long as you use enough trial vectors. However, the better your initial estimates, the quicker and the more easily you'll get a good final result. Also, it's important to make sure that the trial vectors are independent of each other. In the work to follow, we'll be adding the vectors together to get new vectors, something that wouldn't work too well if we had dependent vectors floating around.

Once he'd decided on his trial vectors, Ritz decided to work *as if* these were really the system eigenvectors (which, again, they aren't). But *IF* they actually were the eigenvectors, then Rayleigh's quotient would be stationary with respect to perturbations around them. This is the key to Ritz's breakthrough. He chose to treat his trial vectors as if they were the eigenvectors, and thus he proposed to *force* Rayleigh's quotient to be stationary with respect to perturbations about them. Again, this is a crucial point. Rayleigh's quotient is *not* generally stationary with respect to perturbations about some arbitrary set of trial vectors; rather, it's stationary with respect to the *actual* eigenvectors. Ritz's conceptual leap was to realize that if he forced Rayleigh's quotient to be stationary with respect to his trial vectors, it would result in combinations of trial vectors that were as close as possible to the actual eigenvectors.

Since the trial vectors are independent, it is certainly possible to add them together to get new vectors. And it's also true that if we knew what the actual eigenvectors looked like, we could add them up in such a way as to best approximate the eigenvectors. The Rayleigh-Ritz method will figure out for us how to do this, without actually ever knowing what the real eigenvectors are. Pretty good trick, no?

Here's how it's done. We know that after we've appropriately added together our trial vectors to approximate an eigenvector, the resultant vector will look like this:

$$\text{Est} = \sum_{i=1}^{n} b_i \hat{X}_i \tag{6.4.1}$$

I'm really sorry about all the subscripts and superscripts, the hats and tildes, and whatnot. They're necessary, but I know they're a pain to keep track of. That's why I won't introduce yet another one to stand for "the best estimate of the eigenvector that's constructed from the trial vectors." I'll simply call it Est, for "estimated eigenvector." I hope this makes it a little easier to follow. The $\hat{X}_i$ are our trial vectors and the b_i are the coefficients that cause the overall summation to best approximate the eigenvector.

The only tricky part, therefore, is to determine exactly what these b_i are equal to. If we know that, then we're done, since we'd just use (6.4.1) to find Est. As mentioned, it's the process of forcing Rayleigh's quotient to be stationary that will give us these b_i's.

If we write out Rayleigh's quotient in terms of our estimated eigenvector, we'll have

$$R_Q(\text{Est}) = \frac{\text{Est}^T[K]\text{Est}}{\text{Est}^T[M]\text{Est}} \tag{6.4.2}$$

However, if we look closely at what Est is, we see that the real unknowns are the b_i's. So, since the only real unknowns in Rayleigh's quotient are the b_i's, we can indicate this by writing (6.4.2) as

$$R_Q(\text{Est}) = R_Q(b_i) \tag{6.4.3}$$

Equation (6.4.3) states that the unknown quantities are the b_i's, nothing more. Having said this, we'll refrain from writing the b_i dependency in explicitly, since this would make the following derivations more cluttered than they already are. Just keep in mind that when you see R_Q, this means the Rayleigh's quotient that depends on all the unknown b_i's, as well as the particular $\hat{X}_i$ we've chosen (which are already known).

It's now time to make Rayleigh's quotient stationary. Recall that the definition of stationarity is that the first partial derivatives with respect to each of the independent variables must be zero. This is exactly what we saw in Figures 6.5–6.9. We looked at Rayleigh's quotient around each of the eigenvectors and saw that as we varied our trial vector in the direction of another eigenvector, holding the others fixed, the value of Rayleigh's quotient didn't change. This is precisely what is meant by the first partial derivatives being equal to zero. We therefore require

$$\left.\frac{\partial R_Q}{\partial b_i}\right|_{\text{all } b_i=0} = 0, \quad i = 1, 2, \ldots, n \tag{6.4.4}$$

for every b_i.

Since both the numerator and denominator of Rayleigh's quotient depend on the b_i's, we'll have to use the chain rule to perform this differentiation. If we write Rayleigh's quotient as

$$R_Q = \frac{\text{Num}}{\text{Den}} \tag{6.4.5}$$

where $\text{Num} = \text{Est}^T[K]\text{Est}$ and $\text{Den} = \text{Est}^T[M]\text{Est}$, then

$$\frac{\partial R_Q}{\partial b_i} = \frac{\partial\left[\text{Num}\,(\text{Den})^{-1}\right]}{\partial b_i} = \frac{-\text{Num}\frac{\partial \text{Den}}{\partial b_i} + \text{Den}\frac{\partial \text{Num}}{\partial b_i}}{\text{Den}^2} = 0 \tag{6.4.6}$$

The value of Den is always positive, since it involves the positive masses of the system and quadratic (positive) combinations of the trial vectors. Consequently, we can satisfy (6.4.6) only if

$$-\text{Num}\frac{\partial \text{Den}}{\partial b_i} + \text{Den}\frac{\partial \text{Num}}{\partial b_i} = 0 \tag{6.4.7}$$

or if, after dividing by Den,

$$\frac{\partial \text{Num}}{\partial b_i} - R_Q\frac{\partial \text{Den}}{\partial b_i} = 0 \tag{6.4.8}$$

Now we have to figure out what partial differentiation of the numerator and denominator gives us. If we focus on the first term, we've got

$$\begin{aligned}
\frac{\partial \text{Num}}{\partial b_i} &= \frac{\text{Est}^T[K]\text{Est}}{\partial b_i} \\
&= \frac{\partial}{\partial b_i}\left[(b_1\hat{X}_1^T + b_2\hat{X}_2^T + \cdots)[K](b_1\hat{X}_1 + b_2\hat{X}_2 + \cdots)\right] \\
&= \frac{\partial}{\partial b_i}[b_1^2\hat{X}_1^T[K]\hat{X}_1 + b_2^2\hat{X}_2^T[K]\hat{X}_2 + \cdots \\
&\quad + b_1b_2\hat{X}_1^T[K]\hat{X}_2 + b_2b_1\hat{X}_2^T[K]\hat{X}_1 \\
&\quad + b_1b_3\hat{X}_1^T[K]\hat{X}_3 + b_3b_1\hat{X}_3^T[K]\hat{X}_1 \\
&\quad + b_2b_3\hat{X}_2^T[K]\hat{X}_3 + b_3b_2\hat{X}_3^T[K]\hat{X}_2 + \cdots]
\end{aligned} \tag{6.4.9}$$

What we see in (6.4.9) is a series of terms involving only one particular eigenvector (the $\hat{X}_i^T[K]\hat{X}_i$ terms) and another series involving all the cross-terms (terms like $\hat{X}_i^T[K]\hat{X}_j$). Since we're differentiating with respect to only one of the b_i at a time, most of the terms will drop out, leaving us with

$$\begin{aligned}
\frac{\partial \text{Num}}{\partial b_i} &= 2b_i\hat{X}_i^T[K]\hat{X}_i \\
&\quad + b_1\hat{X}_1^T[K]\hat{X}_i + b_1\hat{X}_i^T[K]\hat{X}_1 \\
&\quad + b_2\hat{X}_2^T[K]\hat{X}_i + b_2\hat{X}_i^T[K]\hat{X}_2 \\
&\quad + b_3\hat{X}_3^T[K]\hat{X}_i + b_3\hat{X}_i^T[K]\hat{X}_3 \\
&\quad + \cdots
\end{aligned} \tag{6.4.10}$$

This can be made a bit more compact by remembering that $[K] = [K]^T$ and $[M] = [M]^T$. Since the terms $\hat{X}_i^T[K]\hat{X}_j$ are just scalar quantities, they are equal to their own transpose. Thus $\left(\hat{X}_i^T[K]\hat{X}_j\right)^T = \hat{X}_j^T[K]^T\hat{X}_i = \hat{X}_j^T[K]\hat{X}_i$. So we see that each of the members of the cross-product pairs in (6.4.10) are equal to the same value, leading to

$$\begin{aligned}
\frac{\partial \text{Num}}{\partial b_i} &= 2b_i\hat{X}_i^T[K]\hat{X}_i \\
&\quad + 2b_1\hat{X}_1^T[K]\hat{X}_i \\
&\quad + 2b_2\hat{X}_2^T[K]\hat{X}_i \\
&\quad + 2b_3\hat{X}_3^T[K]\hat{X}_i \\
&\quad + \cdots
\end{aligned} \tag{6.4.11}$$

You'll note that there's a pretty clear pattern shaping up in these results. To see this, let's first make a couple of new definitions. Rather than writing out all these $\hat{X}_i^T[K]\hat{X}_j$ terms, we'll say that

$$\mathcal{K}_{ij} \equiv \hat{X}_i^T[K]\hat{X}_j \tag{6.4.12}$$

and

$$\mathcal{M}_{ij} \equiv \hat{X}_i^T[M]\hat{X}_j \tag{6.4.13}$$

By combining (6.4.1) and (6.4.5) and making use of (6.4.12) and (6.4.13), we can write out the denominator and numerator of Rayleigh's quotient in the following form

$$\text{Num} = \sum_{i=1}^{n}\sum_{j=1}^{n} b_i b_j \mathcal{K}_{ij} \tag{6.4.14}$$

and

$$\text{Den} = \sum_{i=1}^{n}\sum_{j=1}^{n} b_i b_j \mathcal{M}_{ij} \tag{6.4.15}$$

Written like this, it is clear that Rayleigh's quotient depends only upon the b_i, all the $\mathcal{K}_{ij}$'s and $\mathcal{M}_{ij}$'s being known constants.

Using this new $\mathcal{K}_{ij}$ and $\mathcal{M}_{ij}$ notation, we can rewrite (6.4.11) as

$$\frac{\partial \text{Num}}{\partial b_i} = 2(b_1\mathcal{K}_{i1} + b_2\mathcal{K}_{i2} + b_3\mathcal{K}_{i3} + \cdots) \tag{6.4.16}$$

which can be written as

$$\frac{\partial \text{Num}}{\partial b_i} = 2\begin{bmatrix} \mathcal{K}_{i1} & \mathcal{K}_{i2} & \mathcal{K}_{i3} & \cdots \end{bmatrix}\begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{Bmatrix} \tag{6.4.17}$$

We can now follow exactly the same procedure for $\frac{\partial \text{Den}}{\partial b_i}$. Since the denominator and numerator are identical in form, the only difference being the use of $[M]$ instead of $[K]$, we can immediately realize that

$$\frac{\partial \text{Den}}{\partial b_i} = 2\begin{bmatrix} \mathcal{M}_{i1} & \mathcal{M}_{i2} & \mathcal{M}_{i3} & \cdots \end{bmatrix}\begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{Bmatrix} \tag{6.4.18}$$

So, we can use (6.4.16) and (6.4.18) to write the ith equation (6.4.8) as

$$\begin{bmatrix} \mathcal{K}_{i1} & \mathcal{K}_{i2} & \mathcal{K}_{i3} & \cdots \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{Bmatrix} - R_Q \begin{bmatrix} \mathcal{M}_{i1} & \mathcal{M}_{i2} & \mathcal{M}_{i3} & \cdots \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \end{Bmatrix} = 0 \tag{6.4.19}$$

This is obviously true for any i, and so we can write out all n equations by filling in all the other rows of $[\mathcal{M}]$ and $[\mathcal{K}]$:

$$\begin{bmatrix} \mathcal{K}_{11} & \mathcal{K}_{12} & \vdots \\ \mathcal{K}_{12} & \mathcal{K}_{22} & \vdots \\ \cdots & \cdots & \mathcal{K}_{nn} \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix} - R_Q \begin{bmatrix} \mathcal{M}_{11} & \mathcal{M}_{12} & \vdots \\ \mathcal{M}_{12} & \mathcal{M}_{22} & \vdots \\ \cdots & \cdots & \mathcal{M}_{nn} \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{Bmatrix} \tag{6.4.20}$$

or, more compactly,

$$[\mathcal{K}]B - R_Q[\mathcal{M}]B = 0 \tag{6.4.21}$$

What does this look like? An eigenvalue problem! (What else?) If we realize that R_Q is an unknown scalar, and the elements of B are also unknown, then we see we've got $n + 1$ unknowns and n equations—just like all of our previous eigenvalue problems. If we further realize that R_Q is our approximation to the square of the system's natural frequencies, then (6.4.21) looks exactly like our old friend $[M]\ddot{X} + [K]X = 0$, since the eigenvalue form of this problem is given by $[K]X_i - \omega_i^2[M]X_i = 0$.

Now, let's think a bit about what (6.4.21) is telling us. If we use only one trial vector, then (6.4.21) becomes a scalar equation. In fact, it becomes exactly equivalent to the Rayleigh's quotient approach. If we add another trial vector, then the matrices in (6.4.21) are 2×2. Three trial vectors makes the problem three-dimensional. And so on. What we've got is an eigenvalue problem that has dimensions equal to the number of trial vectors we're using. Let's assume that we want to use three trial vectors. Then, when we solve the associated eigenvalue problem, we'll find three values for the system's natural frequencies (corresponding to the three roots of the characteristic equation for the problem). For each of these solutions, we'll have an associated eigenvector. Note that this is not a *system* eigenvector. Rather, it tells us what the b_i's should be to best approximate a system eigenvector by means of our trial functions. The following examples will illustrate this point.

Example 6.3

Problem We'll start by looking at a simple example, one that's very easy to analyze analytically and will therefore allow us to compare the Rayleigh-Ritz approach against known results. The system we'll consider is shown in Figure 6.13 and is simply that well-used example, a serial chain of masses and springs. For this example we'll let all the springs constants be equal to 1 N/m and all the masses be equal to 1 kg.

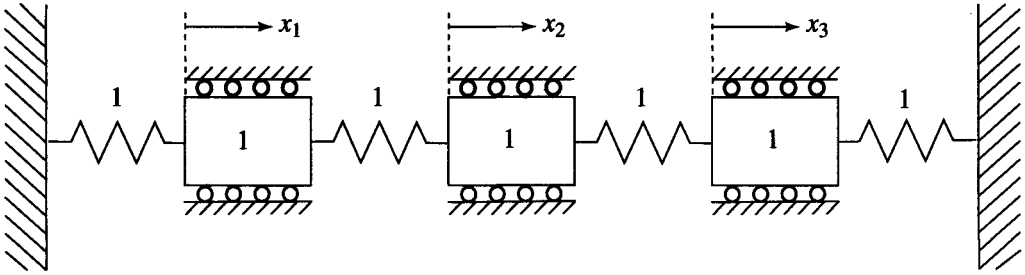

Figure 6.13 3 DOF spring-mass system, spatially symmetric masses, and springs

Solution We know that the eigenvalue problem is given by

$$\left[[K] - \omega^2[M]\right] X = \{0\}$$

and has solutions $\omega_1^2 = .5858\,(\text{rad/s})^2$, $\omega_2^2 = 2.000\,(\text{rad/s})^2$, and $\omega_3^2 = 3.4142\,(\text{rad/s})^2$, with associated eigenvectors

$$X_1 = \begin{Bmatrix} 1 \\ \sqrt{2} \\ 1 \end{Bmatrix}, \quad X_2 = \begin{Bmatrix} 1 \\ 0 \\ -1 \end{Bmatrix}, \quad X_3 = \begin{Bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{Bmatrix} \tag{6.4.22}$$

Notice that these eigenvectors are somewhat special owing to the system's configuration. Since the system has identical masses and identical springs, it will produce eigenvectors that are symmetric with respect to the midpoint of the vector or antisymmetric ones. Thus X_2, an antisymmetric eigenvector, is fundamentally different from X_1 and X_3, which are symmetric. We'll see that this will affect our Rayleigh-Ritz results.

To begin, we'll use

$$\hat{X}_1 = \begin{Bmatrix} 1 \\ 2 \\ 1 \end{Bmatrix}$$

as our initial trial vector. If we calculate $\mathcal{M}_{11}$ and $\mathcal{K}_{11}$ as suggested in (6.4.12) and (6.4.13) we'll find that

$$\mathcal{M}_{11} = 6$$

and

$$\mathcal{K}_{11} = 4.$$

The Rayleigh-Ritz equation is then

$$6 - 4R_Q = 0$$

or $R_Q = .667\,(\text{rad/s})^2$. Since our estimated trial vector differed quite a bit from the actual eigenvector, we obtained only a reasonably accurate eigenvalue estimate for the actual value of $.5858\,(\text{rad/s})^2$. We'll now move on to an analysis featuring two trial vectors.

Example 6.4

Problem Use two trial vectors to analyze the problem of Example 6.3.

Solution This time, we'll say that

$$\hat{X}_1 = \begin{Bmatrix} 1 \\ 2 \\ 1 \end{Bmatrix} \quad \text{and} \quad \hat{X}_2 = \begin{Bmatrix} 1 \\ 0 \\ -1 \end{Bmatrix}$$

This case is special in two ways. For one thing, we've added an antisymmetric vector to our trial set, which formerly consisted of only a symmetric vector. Since the original system is symmetric, it can be broken into two simpler parts, the symmetric parts being independent of the antisymmetric ones, and so it would seem that adding this vector won't do us any good. Since the actual first eigenvector is symmetric, adding an antisymmetric correction by means of the second trial vector shouldn't improve the estimate.

In addition, this second vector happens to be equal to the second eigenvector. Therefore there is no reason for the Rayleigh-Ritz method to add anything to it; it's already as close as it can get to the correct vector. In the same way, adding it to the first vector shouldn't improve the estimate for the first natural frequency because the eigenvectors are independent of each other. And because all the eigenvectors are fundamentally independent of one another, if you're trying to approximate the first eigenvector, adding other eigenvectors to your approximation can't bring you any closer.

Having said all that, let's see what we actually get. Our $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices are

$$[\mathcal{M}] = \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad [\mathcal{K}] = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

and so our eigenvalue problem becomes

$$\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} - R_Q \begin{bmatrix} 6 & 0 \\ 0 & 2 \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \{0\}$$

Notice that there are no off-diagonal terms in this set of equations. As we saw in Chapter 4, this means that the equations are completely uncoupled. Thus what we've got are essentially two individual equations, neither of which influences the other. Solving them gives us

$$R_{Q1} = .\bar{6}\ (\text{rad/s})^2, \quad B_1 = \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} \quad \text{and} \quad R_{Q2} = 2.0\ (\text{rad/s})^2, \quad B_2 = \begin{Bmatrix} 0 \\ 1 \end{Bmatrix}$$

Note that we are using B_i to denote the vector of b_i's associated with R_{Q_i} and that the B_i's components are used to calculate our estimate of the system's eigenvectors. As we expected, the addition of $\hat{X}_2$ didn't alter Est_1 in the slightest, as indicated by the fact that the second component is equal to zero. Thus

$$\text{Est}_1 = b_1\hat{X}_1 + b_2\hat{X}_2 = 1 \begin{Bmatrix} 1 \\ 2 \\ 1 \end{Bmatrix} + 0 \begin{Bmatrix} 1 \\ 0 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 2 \\ 1 \end{Bmatrix}$$

Similarly, the estimate for the second natural frequency and eigenvector is unaffected by $\hat{X}_1$.

Example 6.5

Problem Repeat Example 6.4 with different trial vectors.

Solution Having seen that the choice of trial vectors in Example 6.4 didn't induce any improvement in the eigenvalue estimate, let's see how a different choice fares. For this example we'll use two dissimilar vectors that are neither symmetric nor antisymmetric. Hopefully the Rayleigh-Ritz procedure will use them to create approximations to the actual eigenvalues and eigenvectors.

The two new trial vectors are

$$\hat{X}_1 = \begin{Bmatrix} 1 \\ 3 \\ 2 \end{Bmatrix} \quad \text{and} \quad \hat{X}_2 = \begin{Bmatrix} 2 \\ 1 \\ -1 \end{Bmatrix}$$

We can use these new values for our trial vectors to calculate $[\mathcal{K}]$ and $[\mathcal{M}]$, obtaining

$$\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} - R_Q \begin{bmatrix} 14 & 3 \\ 3 & 6 \end{bmatrix} \begin{Bmatrix} b_1 \\ b_2 \end{Bmatrix} = \{0\} \tag{6.4.23}$$

If we divide $\mathcal{K}_{11}$ by $\mathcal{M}_{11}$ we get .7143 (rad/s)^2 as our single vector estimate for the square of the first natural frequency. You'll note that this is worse than our previous single vector estimate of $.\bar{6}$ (rad/s)^2. This makes sense because this trial vector is even further from the first eigenvector than the original trial vector was. If we calculate the solution to (6.4.23) we'll obtain

$$R_{Q_1} = .\bar{6}\ (\text{rad/s})^2, \quad B_1 = \begin{Bmatrix} 3 \\ 1 \end{Bmatrix} \quad \text{and} \quad R_{Q_2} = 2.0\ (\text{rad/s})^2, \quad B_2 = \begin{Bmatrix} -1 \\ 3 \end{Bmatrix}$$

Interesting. We've obtained the exact same estimates as last time for the natural frequencies. However the associated eigenvectors have changed. Let's take a look at what our new estimated eigenvectors are. The first is

$$\text{Est}_1 = b_1\hat{X}_1 + b_2\hat{X}_2 = 3\begin{Bmatrix} 1 \\ 3 \\ 2 \end{Bmatrix} + 1\begin{Bmatrix} 2 \\ 1 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 10 \\ 5 \end{Bmatrix}$$

This is simply five times our old first trial vector! And since constant multiples make no difference to an eigenvector, we can just as well say it is our old one. The second is given by

$$\text{Est}_2 = -1\begin{Bmatrix} 1 \\ 3 \\ 2 \end{Bmatrix} + 3\begin{Bmatrix} 2 \\ 1 \\ -1 \end{Bmatrix} = \begin{Bmatrix} 5 \\ 0 \\ -5 \end{Bmatrix}$$

which is also five times our old second trial vector. So the new set of two trial vectors must actually be just a linear combination of the old set. This is worth remembering. If you've got a set of trial vectors and another set that are just combinations of the first, then your new frequency estimates will just be the same as the old ones. Along with this goes the fact that your estimated eigenvectors will also remain unchanged. The b_i values will change, of course, since the trial vectors have been altered. But when you examine the final estimated eigenvectors that are indicated by the b_i, you'll see that you get the same answers.

We can gain some more insight into this by examining the $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices and recalling our knowledge of normal coordinates. From Chapter 4 we know the following:

- The normal coordinates are those that decouple our equations of motion.
- The decoupled mass and stiffness matrices are diagonal.
- The normal coordinates are always linear combinations of our original coordinates.

Well, we can view our new set of $\hat{X}_i$'s as our original coordinate set. The Rayleigh-Ritz method takes these vectors and combines them so that Rayleigh's quotient is stationary about them. There's only *one* set of vectors that will satisfy this property of stationarity, and we'll know we have this set because the $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices associated with them will be diagonal. Thus the resultant eigenvectors B_i will be unitary (ones on the diagonal with everything else equal to zero), which will mean that we can't improve the trial vectors by combining them; they're optimal as they are.

With this in mind, take a look at the $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices of our first trial set. They are already in diagonal form. Thus any linear combination of them, when run through the Rayleigh-Ritz method, will simply reproduce them.

It's important to note that even though these matrices are diagonal, they aren't *necessarily* associated with the best possible results. The best would be found if the vectors were exactly equal to the actual eigenvectors. In this case the $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices would be diagonal *and* the eigenvalue estimates would be exact (which they aren't yet for our examples). These are the two ways by which we can obtain completely uncoupled $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices—by using the actual eigenvectors or by choosing (or calculating) the combination of any original trial set that renders Rayleigh quotient stationary.

Having seen what happened with our first two trial sets, let's try a third.

Example 6.6

Problem Repeat Example 6.4 with different trial vectors.

Solution This time, we'll use

$$\hat{X}_1 = \begin{Bmatrix} 4 \\ 3 \\ 0 \end{Bmatrix} \quad \text{and} \quad \hat{X}_2 = \begin{Bmatrix} 0 \\ 2 \\ 4 \end{Bmatrix}$$

The associated $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices are

$$[\mathcal{K}] = \begin{bmatrix} 26 & -8 \\ -8 & 24 \end{bmatrix} \quad \text{and} \quad [\mathcal{M}] = \begin{bmatrix} 25 & 6 \\ 6 & 20 \end{bmatrix}$$

Just using the first trial vector gives us

$$R_Q = \frac{\mathcal{K}_{11}}{\mathcal{M}_{11}} = 1.04 \text{ (rad/s)}^2$$

This is not terribly close to the actual value of .5858 rad/s^2 (having an error of 78%). Let's see how adding the second trial vector helps. If we go ahead and solve the 2×2 eigenvalue problem, we'll find

$$R_{Q_1} = .5961 \text{ (rad/s)}^2, \quad B_1 = \begin{Bmatrix} .2500 \\ .2396 \end{Bmatrix}$$

$$R_{Q_2} = 2.0246 \text{ (rad/s)}^2, \quad B_2 = \begin{Bmatrix} .2500 \\ -.3054 \end{Bmatrix} \tag{6.4.24}$$

These are more typical results. Our original $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices were not decoupled; thus our trial vectors were improved by the Rayleigh-Ritz procedure. Unlike the last time, neither of the frequency estimates was exact, both were simply approximations to the actual eigenvalues. At .5961 (rad/s)^2, our estimate for ω_1^2 is greatly improved from what we had using just our first trial vector. Our original 78% error has been reduced to 2%. To find the associated eigenvector estimate, we need to add .2500 times the first trial vector to .2396 times the second, that is, $B_1(1, 1) * \hat{X}_1 + B_1(2, 1) * \hat{X}_2$. If we do this, we'll obtain

$$\text{Est}_1 = \begin{Bmatrix} 1.000 \\ 1.229 \\ .959 \end{Bmatrix}$$

Clearly, this is a lot closer to the actual eigenvector (shown in 6.4.22) than the initial trial vector was. Our first trial vector was $\{4 \quad 3 \quad 0\}^T$, a crummy guess. By appropriately adding a judicious amount of the second trial vector ($\{0 \quad 2 \quad 4\}^T$), we came up with $\{1.000 \quad 1.229 \quad .959\}^T$, a really good estimate. Similarly, neither initial vector is close to the system's second eigenvector, but by subtracting .3054 times the second trial vector from .2500 times the first, we get $\{1.000 \quad .139 \quad -1.222\}^T$, which is quite close to the actual eigenvector. Furthermore, the error in the estimate of ω_2^2 is only 1%.

We've seen in the preceding example that the Rayleigh-Ritz approach has done an excellent job. We fed in some trial vectors and got out good estimates of the actual system solutions. Of course, the next step is to ask what happens if we add a third trial vector. Well, if we've got three independent, 3×1 vectors to work with, we know that we can create *any* 3×1 vector we might wish to set up; any three independent trial vectors will span the system's configuration space. Therefore, if you give the Rayleigh-Ritz formulation three independent vectors and ask it for the combination that best approximates the actual solution, it will give you the actual solution. That's only fair, since at this point you're doing just as much work as you would have had to do to solve the original problem. Remember, the point of the Rayleigh-Ritz approach is to save time and get good estimates of the first few natural frequencies and eigenfunctions of a continuous system, one that has an infinite number of modes. In this section we restricted the problem to a finite-dimensional one to better understand the method. If the original problem has order n, then to find the natural frequencies and eigenvectors, we need to solve an nth-order eigenvalue problem. By going ahead and using n trial functions, we'll again get an nth-order eigenvalue problem to solve—i.e., no savings. But that's okay because we'd never use the method in this way. The whole point is to use it when you want only a few of the modes, not all of them.

Example 6.7

Problem Add another trial function to Example 6.6.

Solution Just to demonstrate that we do indeed get the precise answer back if we expand our trial function set, we'll again do a Rayleigh-Ritz analysis, this time using

$$\hat{X}_1 = \begin{Bmatrix} 4 \\ 3 \\ 0 \end{Bmatrix}, \quad \hat{X}_2 = \begin{Bmatrix} 0 \\ 2 \\ 4 \end{Bmatrix}, \quad \text{and} \quad \hat{X}_3 = \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix}$$

For these three trial vectors we'll find that

$$[\mathcal{K}] = \begin{bmatrix} 26 & -8 & 4 \\ -8 & 24 & 4 \\ 4 & 4 & 2 \end{bmatrix} \quad \text{and} \quad [\mathcal{M}] = \begin{bmatrix} 25 & 6 & 7 \\ 6 & 20 & 6 \\ 7 & 6 & 3 \end{bmatrix}$$

The full solutions to the Rayleigh-Ritz problem using these matrices are

$$\begin{aligned} R_{Q_1} &= .5858 \text{ (rad/s)}^2 \\ R_{Q_2} &= 2.0000 \text{ (rad/s)}^2 \\ R_{Q_3} &= 3.4142 \text{ (rad/s)}^2 \end{aligned} \tag{6.4.25}$$

with the associated B_i vectors are

$$B_1 = \begin{Bmatrix} .4142 \\ .4142 \\ -.6569 \end{Bmatrix}, \quad B_2 = \begin{Bmatrix} .0 \\ -.5 \\ 1.0 \end{Bmatrix}, \quad \text{and} \quad B_3 = \begin{Bmatrix} -2.4142 \\ -2.4142 \\ 10.6569 \end{Bmatrix}$$

As predicted, these natural frequency estimates are exact. Furthermore, if you construct the first estimated eigenvector you'll see that it's

$$\text{Est}_1 = .4142 \begin{Bmatrix} 4 \\ 3 \\ 0 \end{Bmatrix} + .4142 \begin{Bmatrix} 0 \\ 2 \\ 4 \end{Bmatrix} + -.6569 \begin{Bmatrix} 1 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 1 \\ \sqrt{2} \\ 1 \end{Bmatrix}$$

i.e., the actual first eigenvector. Similarly, Est_2 and Est_3 also give us the exact second and third eigenvectors.

Example 6.8

Problem Example 6.7 gave us a good insight into how the Rayleigh-Ritz method functions. Now we'll try a more complicated problem, the solutions of which will more closely resemble "real-world" results. The system to be analyzed is shown in Figure 6.14. You'll note that it is again a chain of springs and masses. This time, however, it isn't symmetric. All the springs are the same, but the first mass is half the mass of the others.

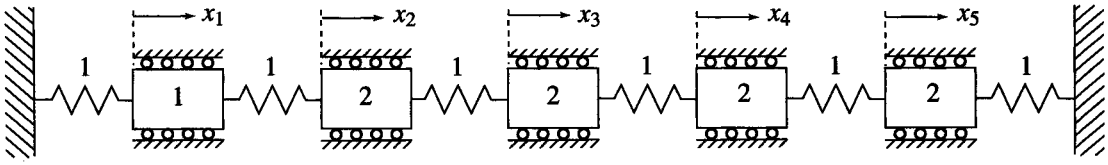

Figure 6.14 5 DOF spring-mass system, spatially nonsymmetric masses, and springs

Solution The $[K]$ and $[M]$ matrices are given by

$$[K] = \begin{bmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{bmatrix}$$

and

$$[M] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

Our trial functions will be

$$\hat{X}_1 = \begin{Bmatrix} 2 \\ 3 \\ 4 \\ 3 \\ 1 \end{Bmatrix}, \quad \hat{X}_2 = \begin{Bmatrix} 2 \\ 2 \\ 0 \\ -1 \\ -1 \end{Bmatrix}, \quad \hat{X}_3 = \begin{Bmatrix} 2 \\ 0 \\ -1 \\ 0 \\ 1 \end{Bmatrix}, \quad \hat{X}_4 = \begin{Bmatrix} 2 \\ -1 \\ 0 \\ 1 \\ -1 \end{Bmatrix},$$

and

$$\hat{X}_5 = \begin{Bmatrix} .5 \\ -.4 \\ 1. \\ -.3 \\ .4 \end{Bmatrix}$$

Based on these trial vectors and the given $[M]$ and $[K]$ matrices, the $[\mathcal{K}]$ and $[\mathcal{M}]$ matrices are found to be

$$[\mathcal{K}] = \begin{bmatrix} 12 & 2 & -1 & 4 & 1.8 \\ 2 & 10 & 4 & 2 & -.9 \\ -1 & 4 & 12 & 7 & 1.2 \\ 4 & 2 & 7 & 20 & 2 \\ 1.8 & -.9 & 1.2 & 2 & 5.36 \end{bmatrix}$$

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and

$$[\mathcal{M}] = \begin{bmatrix} 74 & 8 & -2 & 2 & 5.6 \\ 8 & 16 & 2 & 0 & -.8 \\ -2 & 2 & 8 & 2 & -.2 \\ 2 & 0 & 2 & 10 & .4 \\ 5.6 & -.8 & -.2 & .4 & 3.07 \end{bmatrix}$$

If we now go through and solve all the possible Rayleigh-Ritz problems (one trial vector, two trial vectors, etc.) we'll finally obtain the estimates shown in Table 6.1.

Here we see that every time a new vector is added to our trial set, the estimates get better. We start with the Rayleigh's quotient answer of .1622. Upon adding a second vector, this estimate goes to .1612, and we also get an estimate of the square of the second natural frequency (.6423). Note that when a third trial vector is added, the first estimate drops by quite a bit more than it did when we went from one vector to two (.1612 down to .1546, vs. 1622 to .1612). The reason for this is that the second trial vector didn't really do much for us in terms of improving our approximation of the first eigenvector. However the first trial vector plus the third allow us to get a much better approximation, thus the bigger improvement in our estimate. The answers for five trial vectors are exact, since the original system only involves five eigenvectors, just as we saw in Example 6.7.

It's time for a brief recap before moving on to continuous systems. Let's say for the moment that the $[M]$ and $[K]$ matrices for our actual system are 10×10. The easiest attempt at finding the fundamental frequency involves a single equation. This is the Rayleigh's quotient approach. If we decide to use more than one trial vector, we've got to use the Rayleigh-Ritz approach. As we add more and more trial vectors, our estimates for the system natural frequencies and eigenvectors become better and better. Each time we're solving an eigenvalue problem whose size is equal to the number of trial vectors. If we go all the way up to 10 independent trial vectors, we're solving a problem that's every bit as hard as the actual problem we were trying to approximate.

We can also add a few more observations. First, you probably noticed that our Rayleigh-Ritz estimates always went down or remained unchanged as we added trial vectors. This isn't a coincidence. If you think about the problem physically, what the Rayleigh-Ritz method is doing is forcing the system to behave as if it can move only in certain vector directions. For instance, the actual system might involve five masses, as in the preceding example. But by using only three trial vectors, we're constraining the system. We're saying to it, "We don't care how the system

TABLE 6.1 ω_i^2 ESTIMATES FOR 1–5 TRIAL VECTOR (rad/s)^2

	Number of Trial Vectors				
	1	2	3	4	5
R_{Q_1}	.1622	.1612	.1546	.1424	.1393
R_{Q_2}		.6423	.5705	.5688	.5558
R_{Q_3}			1.511	1.176	1.158
R_{Q_4}				2.153	1.733
R_{Q_5}					2.414